

6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

```
clc;
clear;
close all;

fs = 1000;
f = 50;

t = 0:1/fs:0.1;
x = sin(2*pi*f*t);

L = [8 16 32];

colors = {'r','g','m'};

% Quantized Signal Figure
figure;

subplot(4,1,1);
plot(t,x,'b','LineWidth',1.5);
title('Original Sinusoidal Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

for i = 1:3

    xq{i} = round((x+1).*(L(i)-1)/2) ...
        .*(2/(L(i)-1)) - 1;

    e{i} = x - xq{i};
```

```
mse(i) = mean(e{i}.^2);
```

```
subplot(4,1,i+1);
```

```
stairs(t,xq{i},colors{i},'LineWidth',1.2);
```

```
title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
end
```

```
fprintf('MSE for 8 Levels = %f\n',mse(1));
```

```
fprintf('MSE for 16 Levels = %f\n',mse(2));
```

```
fprintf('MSE for 32 Levels = %f\n',mse(3));
```

```
% Quantization Error Figure
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(t,e{1});
```

```
title('Quantization Error (8 Levels)');
```

```
xlabel('Time (s)');
```

```
ylabel('Error');
```

```
grid on;
```

```
subplot(3,1,2);
```

```
plot(t,e{2});
```

```
title('Quantization Error (16 Levels)');
```

```
xlabel('Time (s)');
```

```
ylabel('Error');
```

```
grid on;
```

```
subplot(3,1,3);
```

```
plot(t,e{3});
```

```
title('Quantization Error (32 Levels)');
```

```
xlabel('Time (s)');
```

```
ylabel('Error');
```

```
grid on;
```



